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DAY 4

MACHINE LEARNING BASICS

How Computers Learn From Data Without Being Told the Rules

A Course Guide by AYE Tech Hub

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1. Introduction

In Day 3 you saw *where* AI lives in your daily life — recommendations, voice assistants, maps, fraud detection. Today you will learn *how* all of those systems actually work under the hood. The answer is almost always the same: **Machine Learning (ML)** — the engine that powers about 95% of modern AI applications.

Machine Learning is not magic, and it is not new — the core ideas come from statistics in the 1950s. What is new is the combination of **cheap computing power**, **massive datasets**, and **better algorithms** that, together, allow computers to discover patterns humans cannot see. After today, you will understand what an ML model is, how it learns, and why some predictions are amazingly accurate while others fail.

Learning Objectives

By the end of this lesson you will be able to:

- Explain the difference between traditional programming and machine learning.
- Define the three main types of learning (supervised, unsupervised, reinforcement) and give two real-world examples of each.
- Describe the six steps of the ML workflow from raw data to deployment.
- Recognise overfitting and underfitting and explain how each affects accuracy.
- Choose the right family of algorithm (regression / classification / clustering) for a given business problem.

AI vs ML vs Deep Learning — Quick Map

Three terms often confused:

- **Artificial Intelligence (AI)** — the broad goal of making computers act intelligently.
- **Machine Learning (ML)** — one approach to AI: learning from data instead of hard-coded rules.
- **Deep Learning (DL)** — a powerful subset of ML using multi-layer neural networks. Think of it as Russian dolls: **AI ⊃ ML ⊃ DL**.

2. Traditional Programming vs Machine Learning

To understand ML, start with what came before it. In **traditional programming**, a human expert writes explicit rules. The computer applies those rules to data and produces an answer. This works perfectly for problems where the rules are simple and well-known — calculating tax, sorting a list, controlling a PLC sequence.

But for problems like 'is this email spam?' or 'whose face is in this photo?', writing rules by hand is almost impossible. There are too many variations, too many exceptions. **Machine Learning flips the relationship**: you give the computer data plus correct answers, and the computer figures out the rules itself. The model that comes out can then be applied to new data the computer has never seen.

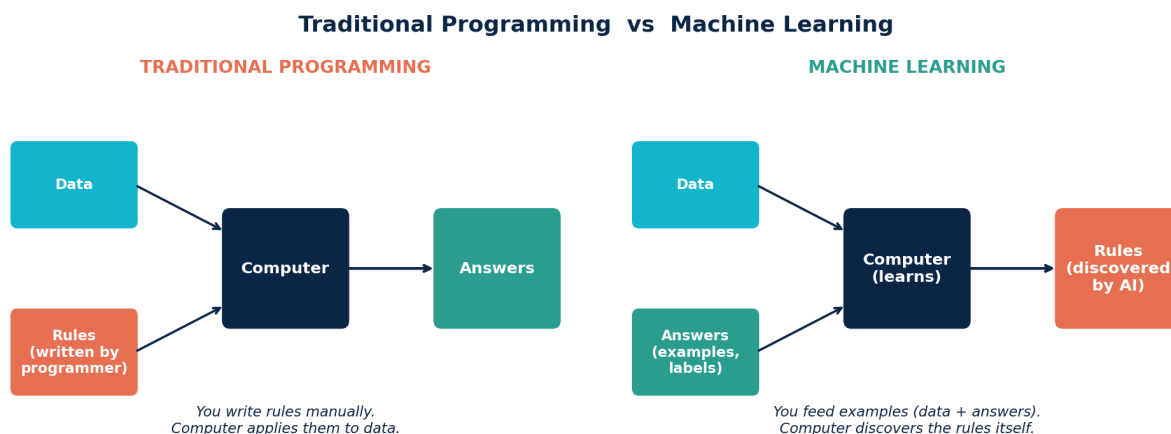


Figure 2.1 — Traditional programming takes data + rules and produces answers. Machine Learning takes data + answers and produces rules.

Aspect	Traditional Programming	Machine Learning
Who writes the rules?	Human programmer	Computer learns rules from data
Input to the computer	Data + Rules	Data + Answers (examples)
Output	Answers	A trained model (the rules)
Best for	Well-defined logic, math, control	Pattern recognition, prediction
Example	PLC ladder logic, tax calculator	Face recognition, fraud detection
What changes when data changes?	Rules must be rewritten manually	Model is retrained automatically

3. The Three Types of Machine Learning

Not all ML problems look the same. Depending on what kind of data you have and what you want to predict, you choose one of three learning approaches. Knowing the difference is the most important conceptual skill in this entire field — every ML project starts by asking 'which type of learning is this?'

The Three Main Types of Machine Learning

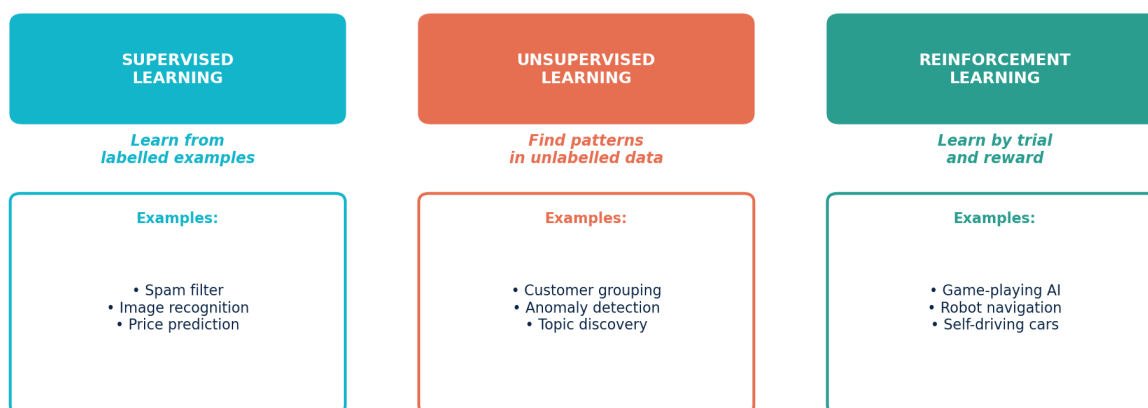


Figure 3.1 — Supervised, unsupervised, and reinforcement learning. Each works on a different kind of data and answers a different kind of question.

Type	What You Give the AI	What the AI Learns	Real Use
Supervised	Data WITH labels (input + correct answer)	To predict the label for new data	Spam filter, image recognition, price prediction
Unsupervised	Data WITHOUT labels	Hidden structure or groupings in the data	Customer segmentation, anomaly detection
Reinforcement	An environment + reward signal	Best actions to maximise reward	Game-playing AI, robot control, trading bots

Real-World Hint

If someone hands you a labelled spreadsheet (columns of input + a column of answers), that is a **supervised learning** problem. If you only have inputs and want to find hidden groups, that is **unsupervised**. If your AI must take actions over time and get rewarded or penalised, that is **reinforcement learning**. 90% of business ML projects are supervised — it is the most common starting point.

4. Supervised Learning in Detail

Supervised learning is the workhorse of modern AI. The name comes from the fact that during training, the AI is 'supervised' by a human-provided correct answer for every example. Once trained, the model can predict the answer for new data it has never seen.

The classic example is a **spam filter**. You collect thousands of emails, each one labelled as 'spam' or 'not spam' by a human. The AI looks at the words, sender, and patterns in each email, and learns which features predict spam. Once trained, it can classify new emails automatically — and the accuracy keeps improving as more labelled examples flow in.

Supervised Learning Example — Building a Spam Filter

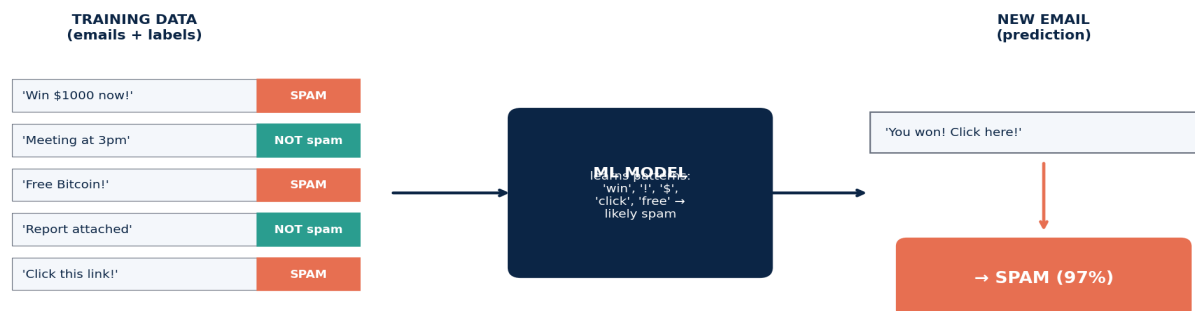


Figure 4.1 — A complete supervised-learning loop: labelled training data → trained model → prediction on a new, unseen email.

Two Sub-Types of Supervised Learning

Sub-Type	What It Predicts	Example Output	Real Example
Classification	A category (label)	'spam' or 'not spam'	Email filter, image labelling, disease diagnosis
Regression	A continuous number	87,500 KRW	House price, electricity demand, sales forecast

Worked Example — Predicting Apartment Price

Imagine you want to predict apartment prices in Gimpo. You collect 1,000 past sales with **features** (size in m², floor, age, distance to subway) and the **label** (actual selling price). A regression model learns the relationship — for example, that each m² adds ~3 million KRW, each year of age subtracts ~500 000 KRW, etc. Once trained, you give it features of a new apartment and it predicts the price.

5. Unsupervised Learning in Detail

Sometimes you have data but no answers — no labels. **Unsupervised learning** finds structure in such data without being told what to look for. The most common technique is **clustering**: grouping similar items together. Other techniques find associations ('people who bought X also bought Y') or detect outliers (anomalies that don't fit any group — useful for fraud detection).

Unsupervised Learning — Finding Hidden Groups in Data

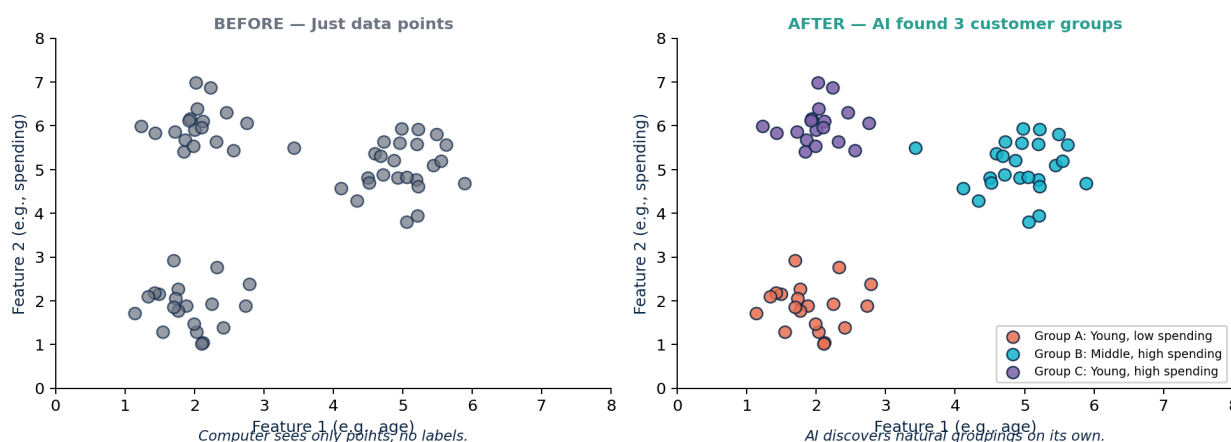


Figure 5.1 — Clustering: the AI takes unlabelled data points (left) and discovers natural groupings on its own (right). Each colour is a group the AI invented.

Real-World Use Cases

Use Case	What the AI Discovers	Business Value
Customer segmentation	Groups of customers with similar buying habits	Targeted marketing, loyalty programs
Anomaly detection	Transactions or events that don't fit any normal pattern	Fraud detection, equipment failure prediction
Topic modelling	Themes that emerge from large text collections	News categorisation, customer feedback analysis
Recommendation	Items that are similar or are bought together	Product recommendations, content discovery
Dimensionality reduction	Hidden simpler structure in complex data	Data visualisation, feature compression

Engineer's Insight

A factory like ASAN Auto Parts can use unsupervised learning to find **abnormal machine behaviour**. Sensors on each press, robot, and conveyor stream temperature, vibration, and current readings 24/7. An anomaly-detection model learns the 'normal' pattern for each machine. When something starts drifting outside normal — bearing wear, loose bolt, motor overheating — it flags the maintenance team **before** the breakdown. This is called predictive maintenance, and it can cut downtime by 30–50%.

6. Reinforcement Learning

Reinforcement learning (RL) is the third paradigm — and the most fascinating. There is no fixed dataset. Instead, an **agent** (the AI) interacts with an **environment** (a game, a road, a market). The agent takes actions, observes the result, and receives a **reward** (positive or negative). Over thousands or millions of trials, it learns which actions lead to the highest reward in each situation.

Reinforcement Learning — Learning by Trial and Reward

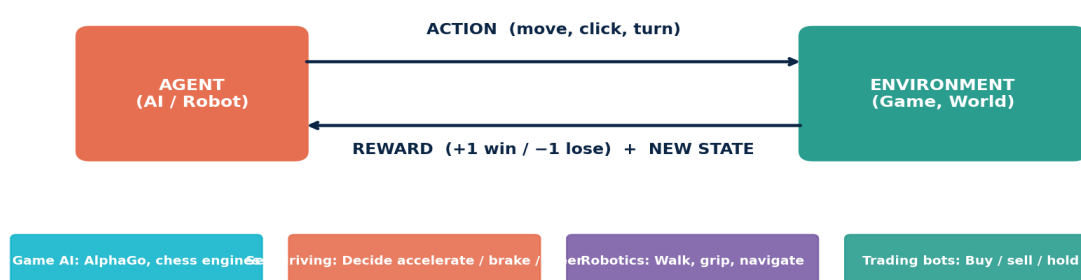


Figure 6.1 — The reinforcement-learning loop. The agent acts, the environment responds with a new state and a reward, and the agent updates its strategy.

Famous Real-World Wins

System	Maker	What It Learned
AlphaGo / AlphaZero	Google DeepMind	Beat world champion at Go, chess, shogi
OpenAI Five	OpenAI	Beat professional Dota 2 teams
Tesla / Waymo self-driving	Tesla, Waymo	Lane changes, merging, traffic decisions
Boston Dynamics Atlas	Boston Dynamics	Walking, running, jumping, recovery from pushes

System	Maker	What It Learned
ChatGPT (RLHF stage)	OpenAI	Reinforcement Learning from Human Feedback to be helpful
Industrial robot arms	Fanuc, ABB, Universal Robots	Pick-and-place, assembly, packing

Why RL Is Hard

Reinforcement learning is powerful but data-hungry. To learn to play Go at superhuman level, AlphaZero played **tens of millions of games against itself**. Real robots cannot do that — every fall costs hardware. So researchers train RL agents in **simulation** first, then transfer the learned behaviour to the real robot. This is called 'sim-to-real' and is an active frontier of robotics today.

7. The Machine Learning Workflow — 6 Steps

Every real ML project — from a personal hobby model to a billion-dollar production system at Samsung or Naver — follows roughly the same six steps. Understanding the workflow lets you scope projects realistically and recognise where most projects actually spend their time: collecting and cleaning data, not training the model.

The Machine Learning Workflow — 6 Steps

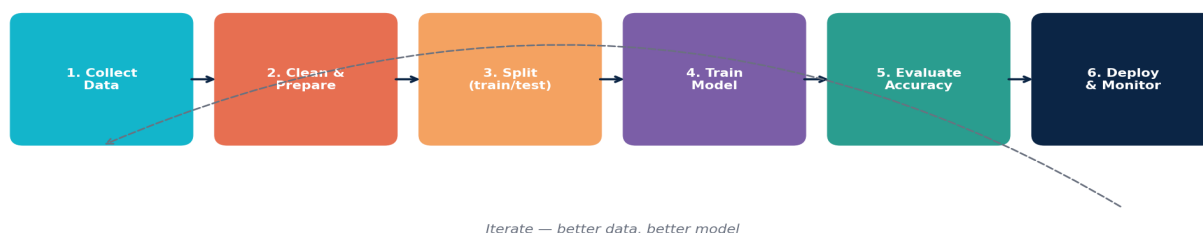


Figure 7.1 — The six-step ML workflow. Steps 1 and 2 (data) typically take 60–80% of total project time, even though step 4 (training) gets all the attention.

Step	What Happens	Time Spent
1. Collect Data	Gather raw data from databases, sensors, files, APIs	High
2. Clean & Prepare	Remove errors, fill missing values, create features	Very High (~50%)
3. Split	Divide data into train (70%), validation (15%), test (15%)	Low
4. Train Model	Choose algorithm, feed training data, adjust parameters	Medium
5. Evaluate	Test on unseen data, measure accuracy, check for bias	Medium
6. Deploy & Monitor	Put model in production; watch performance; retrain when accuracy drops	Continuous

Industry Reality

Data scientists often joke that 80% of their job is cleaning data and only 20% is *actual* machine learning. The popular phrase is '**garbage in, garbage out**' — a brilliant algorithm trained on bad data will produce bad predictions. Always invest more effort in data quality than in choosing exotic algorithms.

8. Overfitting, Underfitting & Accuracy

Once trained, how do you know your model is good? You test it on data it has never seen. Three things can happen:

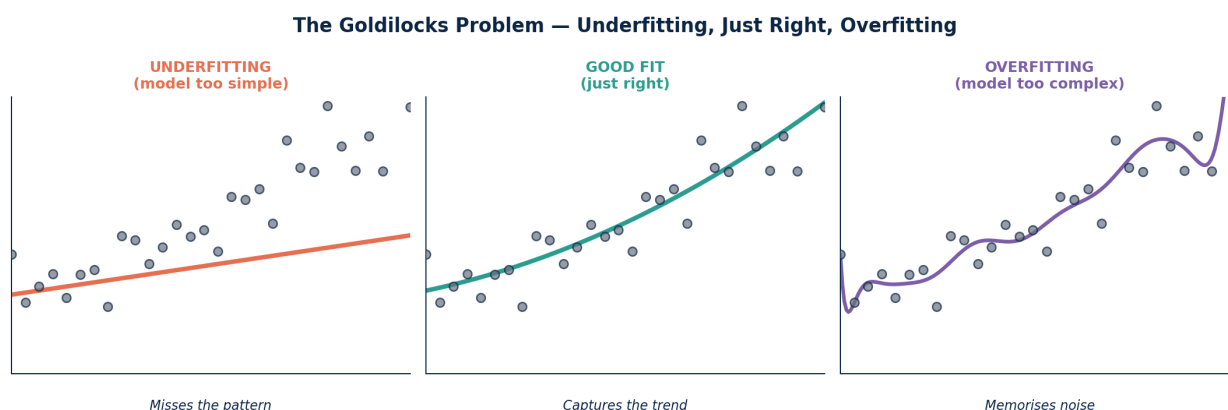


Figure 8.1 — The Goldilocks problem. Left: model too simple, misses the pattern. Centre: just right, captures the trend. Right: model too complex, memorises noise instead of learning the pattern.

Problem	What It Looks Like	Fix
Underfitting	Low accuracy on BOTH training and test data	Use a more powerful model, add more features
Overfitting	High training accuracy, poor test accuracy	More data, simpler model, regularisation
Good Fit	Similar accuracy on training and test	Production ready — deploy and monitor

9. Common Algorithms — Cheat Sheet

Common ML Algorithms — Which One to Use?

PREDICT A NUMBER (Regression)	PREDICT A CATEGORY (Classification)	FIND GROUPS (Clustering)
- Linear Regression - Decision Tree - Random Forest - Neural Network	- Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Neural Network	- K-Means - Hierarchical Clustering - DBSCAN - Gaussian Mixture

Real example: 'Will this customer buy?' → Classification (yes/no)

Figure 9.1 — Which family of algorithm to use depends on what you want to predict: a number, a category, or hidden groups.

10. Summary, Self-Check Quiz & Homework

Lesson Recap

- **Machine Learning** = computers learning rules from data + examples, instead of humans writing rules.
- Three main types: **supervised** (labelled data), **unsupervised** (no labels), **reinforcement** (learn by reward).
- Supervised splits into **classification** (predict a category) and **regression** (predict a number).
- The 6-step ML workflow: collect → clean → split → train → evaluate → deploy. Cleaning data takes the most time.
- **Overfitting** = model memorises training data and fails on new data. **Underfitting** = model is too simple to capture the pattern.
- Choose the algorithm family by what you want to predict: number → regression, category → classification, hidden groups → clustering.

Self-Check Quiz (answers at the bottom)

1. A bank wants to predict if a transaction is fraud. Which type of learning is this?
2. Netflix groups users by viewing taste without knowing their preferences upfront. Which type of learning?
3. An ML model gets 98% accuracy on training data but only 60% on new data. What problem is this?
4. You want to predict tomorrow's electricity demand (a number). Classification or regression?
5. What two ingredients does a supervised-learning system need during training?

Answers: 1) Supervised (classification). 2) Unsupervised (clustering). 3) Overfitting. 4) Regression. 5) Data + labels (correct answers).

Homework Assignment

Choose one ML application you use daily (Netflix, YouTube, your bank app, Naver Maps, or a fitness tracker) and complete the following:

1. Identify which **type of learning** it most likely uses (supervised / unsupervised / reinforcement) and explain why.
2. List three **features** (inputs) the AI probably uses to make decisions for you.
3. What is the **label** or output it predicts?
4. Could it suffer from overfitting? Describe one scenario where its predictions might be wrong because of bias in training data.
5. Suggest one way the company could improve the model.

Submit to awetgknway@gmail.com with subject *AI Day 4 Homework — Your Name*.

Coming Up Tomorrow — Day 5

Deep Learning & Neural Networks — the breakthrough technology that powers ChatGPT, image generation, voice cloning, and self-driving cars. You'll learn what a neural network actually is, how it 'thinks' in layers, and why bigger networks have transformed the field in the past decade.

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